

# Dev Dheeren Bhamaria

Offline-first app developer · BCA student

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## SUMMARY

BCA student who builds free, open-source apps that run entirely on the user's own machine. Shipped a 28-tool PDF suite that works in the browser and as a Windows desktop app, and a GPU-accelerated offline dictation app for Windows. Comfortable across TypeScript, Python, WebAssembly and local AI inference, with a habit of measuring and testing what I ship.

## PROJECTS

### IHatePDF

v1.3.1 · github.com/Burthcer/IHatePDF

TypeScript, React, Vite, Tailwind, pdf-lib, pdf.js, WebAssembly, Electron

- Built 28 PDF tools (edit, merge, split, compress, convert, sign, redact, protect) that run fully client-side in the browser or as a Windows desktop app. No file is ever uploaded.
- Wrote an in-place text editor that retypes a paragraph in the document's own font and removes the original characters instead of covering them.
- Target-size compression: a 35.5 MB scan to 1.95 MB for a 2 MB target; a 500 MB scan compressed to 38.4 MB in 91 s on an 8 GB memory budget.
- Added a memory fail-safe for low-RAM PCs and offline phone sharing over Wi-Fi, hotspot or Bluetooth. 80 automated and end-to-end checks passing; released with a Windows installer.

### Wisperno

github.com/Burthcer/Wisperno

Python 3.12, PySide6 (Qt), CUDA, faster-whisper, llama-cpp-python

- Push-to-talk dictation for Windows: transcribes locally with Whisper large-v3-turbo on CUDA and cleans the text with a quantized Qwen2.5-1.5B. No audio leaves the machine.
- 0.82 s end-to-end latency for 100 words at 113.8 tok/s decode, in about 3.5 GB of VRAM on the default preset.
- Live transcription mode re-decodes the full unconfirmed buffer every 0.6 s at a real-time factor of 0.074 (about 13.5x real time), replacing a sliding-window design that silently dropped content.
- Benchmarked four engine presets and upgraded llama-cpp-python from 0.2.90 to 0.3.26, making every preset 20 to 30% faster with no change in VRAM.

## EDUCATION

### Bachelor of Computer Applications (BCA)

Aug 2024 to 2029 (expected)

D Y Patil Deemed to be University, School of Humanities and Sciences, Navi Mumbai

## SKILLS

### Languages

TypeScript, JavaScript, Python

### Engines and libraries

WebAssembly, pdf-lib, pdf.js, CUDA, faster-whisper, llama-cpp-python

### Interfaces

React, Tailwind, PySide6 (Qt), Electron

### Practice

Offline and air-gapped testing, benchmarking, threat modeling, reverse engineering, Git and GitHub